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PSYCHOENDOCRINE ASPECTS OF MOTHER-INFANT RELATIONSHIPS IN NONHUMAN PRIMATES

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SUMMARY

Studies will be presented which examine the physiological and behavioral responses of squirrel monkeys and rhesus macaques following disruptions of mother-infant relationships. Reliable increases in circulating levels of plasma cortisol occur following separation of the infant from its mother. The presence of familiar conspecifics during the time of separation reduces the pituitary-adrenal response, compared to that elicited by total isolation. Visual access to the mother during separation also ameliorates the plasma cortisol response. However, when infants are separated in the presence of unfamiliar conspecifics, the physiological response is exaggerated compared to animals which are totally isolated.

The behavior expressed by the infant during separation, particularly separation-induced vocalizations, is not concordant with this physiological index of affect. The rate of vocalization produced when the infant has visual access to the mother was higher than when the infant was totally isolated. However, when allowed access to familiar conspecifics, the rate of vocalization was lower than during total isolation, with no vocalization produced while the separated infant was in the unfamiliar social group. The curvilinear relationship between vocalization and the physiological index of arousal has led to a revision of the traditional concept that separation-induced infant vocalization is reflective of distress.

These data support the hypothesis that vocalizations may serve as a coping response that reduces the physiological indices of arousal. Social interaction with familiar conspecifics may serve as a non-vocal coping response (e.g., proximity contact to other monkeys) which also reduces the behavioral and physiological responses to maternal separation.

INTRODUCTION

ALTHOUGH THERE ARE numerous species-specific differences in the quality of mother-infant interaction in nonhuman primates, one general characteristic of the early relationship is its continuity. The infant plays an active role in the maintenance of this dyadic relationship. Contact between the mother and infant monkey, for example, is largely dependent upon the infant' s clinging and following behavior. In terrestrial species such as macaques, shortly after

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birth the infant spends most of its time in ventral/ventral contact with the mother. However, within a short time, the infant makes short forays away from the mother but continues to stay in close proximity to her. The infant gradually spends more and more time away from the mother as it develops; however, it is clear that when the infant is distressed, or there are perturbations in the environment, the infant will actively seek the mother and achieve proximity to her either by clinging or by maintaining a position within arm's reach. This occurs until quite late in development, even while the infant is spending most of its time away from the mother.

In arboreal species, such as the squirrel monkey, the infant assumes a position upon the dorsal surface of the mother almost immediately after birth. It maintains this position for a considerable period of time, until it begins to leave the mother's back, initially briefly, and then for sustained periods of time. During this period, in the squirrel monkey, like the rhesus, the presence of stimuli which may pose a threat or elicit distress results in the infant' s immediately seeking the mother and once again resuming the dorsal clinging position.

Both squirrel monkey and macaque infants emit signals, such as vocalizations, that result in the mother's attempting to achieve proximity with the infant very quickly. The rhesus mother will actively seek the infant and retrieve it vigorously. Squirrel monkey mothers are also active in achieving proximity to their infants; however, they do not retrieve their infants in the usual sense. Under circumstances of stress they will seek the infant and present themselves so that the infant can more easily assume the clinging position. Thus, one of the characteristic responses of human and nonhuman primate mothers and infants under conditions of stress is to achieve proximity and contact.

For most primate species, with a few exceptions (Rosenblum & Kaufman, 1967), the only salient animal in the environment to which the infant will respond is the mother. Most primate mothers appear to respond selectively to the signals emitted by their infants. Separating the primate infant from its mother has been viewed as a highly stressful procedure (Harlow & Harlow, 1962; Kaufman, 1973). The behavioral consequences have been studied extensively (Seay *et al.*, 1962; Hinde & Spencer-Booth, 1967; Kaufman, 1973; Mineka & Suomi, 1978); however, the assumption that separation is stressful to the infant has been based primarily on the behavior of the infant during the period of separation.

The response of infants following separation has been characterized as biphasic, involving an initial "protest response" characterized by increased levels of activity and vocalization, followed by a "depression" in which the infants show decreased activity and vocalization, a hunched over posture, withdrawal, and an increase in self-directed behaviors. The intensity and duration of each phase of the response varies as a function of the separation environment, the species, and the infant's prior experience (Mineka & Suomi, 1978). Variations in the infant's behavioral responses following separation have been used to infer the extent of the distress resulting from loss of the mother (Hinde & Davies, 1972; Chappell & Meier, 1975; Mineka & Suomi, 1978). However, the behavioral response to maternal separation shows considerable variation, and frequently the "biphasic protest and despair response" has not been observed in infant primates (Rosenblum & Kaufman, 1968).

In our laboratory, we have failed, after numerous experiments, to produce behavioral responses indicative of depression in both squirrel monkeys and rhesus macaques. In the experiment with macaques, even after a prolonged two-week separation from the mother, a depression response was not observed (Gunnar *et al.*, 1981). However, there are many methodological differences between this study and others which have shown responses to separation. One of the purposes of this review is to highlight the importance of the separation environment for both the biological and behavioral responses following separation of the infant from its mother.

Over the past several years there has been a marked increase in the number of studies which have investigated biological changes that occur following disruptions of the nonhuman primate

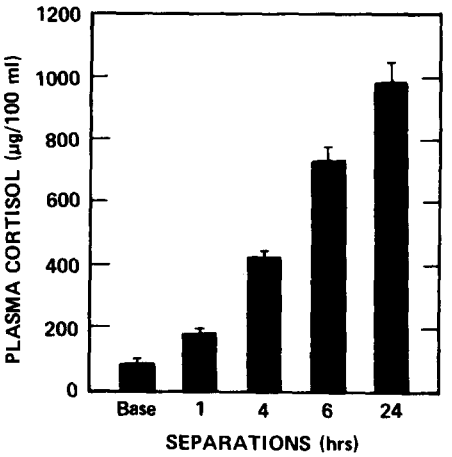


FIG. 1: Infant squirrel monkey plasma cortisol concentrations (µg/100 ml) under a nondisturbed home condition (BASE) and following a 1-, 4-, 6- or 24-hr total separation from the mother. Shown are means ± SEM.

mother-infant relationship. Many of the recent studies have examined not only the behavioral consequences of separation, but also several biological parameters that change in response to separation (Reite *et al.*, 1978; Coe *et al.*, 1985). Our approach to studying the biological consequences of separation is to examine the effects of separation upon the activity of the hypothalamic pituitary-adrenal system. There are several reasons for choosing to investigate this particular biological system. Our laboratory has had a sustained interest in attempting to specify the psychological variables which either activate or inhibit (Hennessy & Levine, 1979) the pituitary-adrenal system.

When we first began our studies on the endocrine response to separation, we believed that we were investigating the biological sequelae of attachment (Bowlby, 1969; 1973). The infant's behavioral responses following loss of the attachment figure have been described in numerous studies (Mineka & Suomi, 1978). Because the loss of the attachment figure is considered stressful, and the hormones secreted by the hypothalamic pituitary-adrenal system (CRF, ACTH, cortisol) fall into the general category of stress hormones, it is reasonable to expect that plasma cortisol should change with the loss of the attachment figure.

Although our primary goal was to examine the biological sequalae that occur as a consequence of loss of the attachment figure, our use of plasma cortisol as a dependent measure has resulted in an interpretation which does not rely on attachment as its major theoretical construct, but rather on the concepts of stress and coping. The data accumulated over what is now almost a decade of research indicate that the response of the pituitary-adrenal system is robust, consistent, and sensitive to conditions of separation.

Endocrine Response to Separation

Our first nonhuman primate study demonstrated that both squirrel monkey infants and their mothers displayed significant elevations in plasma cortisol concentrations to a brief (30 min) period of separation (Mendoza *et al.*, 1978). Although this study was based on a very brief period of separation, during the ensuing years we have demonstrated that the endocrine response to separation in the infant squirrel monkey increases as the length of separation increases, up to 24 hr. Thus, there is an incremental rise in circulating levels of plasma cortisol,

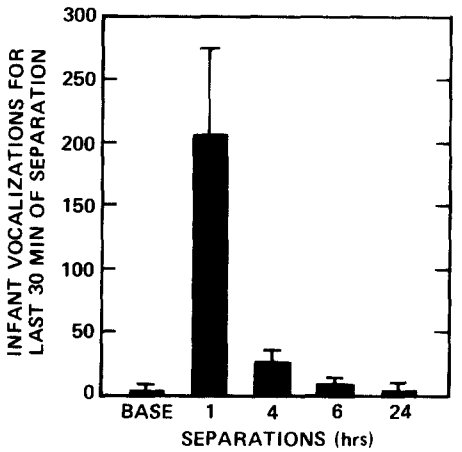


FIG. 2: Infant squirrel monkey isolation vocalizations during a nondisturbed home condition (BASE) and during the last 30 min of a 1-, 4-, 6- or 24-hr total separation from the mother. Shown are means ± SEM.

with a peak occurring approximately 24 hr following separation (Fig. 1). Although we have not studied periods longer than 24 hr in squirrel monkeys less than six months of age, we were able to study longer periods of separation in subsequent studies by taking advantage of a naturally occurring event, weaning, which results in a prolonged and permanent separation from the mother. Our routine laboratory procedure is to remove the infant squirrel monkey from its mother at six to eight months of age, in order to promote subsequent breeding. The initial period following weaning, therefore, represents a period of extended separation. Under these circumstances, plasma cortisol remains elevated for a considerable length of time (i.e., weeks) before returning to baseline.

If we examine the relationship between vocalization and cortisol concentration, we find, however, that there is an apparent inverse relationship. As we will demonstrate later, both the qualitative and quantitative aspects of vocalization tend to vary with conditions of separation. Vocalizations tend to show a decrease following separation, with a peak in vocalization occurring within one hour after separation (Fig. 2). By 24 hr, when plasma cortisol is at its highest, vocalizations are minimal or nonexistent. Thus, high levels of vocalization are associated with reduced cortisol concentrations. These results raise specific questions concerning the function of vocalization following separation and the value of the vocalization response as an index of distress (Panksepp *et al.*, 1985).

Coping Theory of the Separation Response

Suomi *et al.* (1983) stated, "There is now considerable evidence that when most human and nonhuman primates are forcibly separated from their mothers they become agitated and distressed, a response first termed protest by Robertson and Bowlby (1952)." The concept of protest is defined in terms of vocalization and movement. However, the term clearly implies a set of affective responses that underlie its operational definition. Here one should expect high levels of vocalization would be positively associated with high levels of pituitary-adrenal activity. This is based on the assumption that vocalization is equivalent to distress. We would like to propose a different hypothesis concerning the meaning of vocalization rate and quality

that deals specifically with the adaptive value of infant vocalizations following separation. As indicated above, we have attempted to incorporate certain aspects of the mother-infant relationship, particularly the behavioral events that occur following separation, into coping theory (Levine, 1983). We have hypothesized that under conditions of environmental perturbation, proximity and contact with the mother result in reduction of the infant' s arousal levels. One manifestation of the infant's reduced arousal is the reduction or elimination of its plasma cortisol response to these environmental disturbances (Hennessy & Levine, 1979). Proximity behavior that results in arousal reduction involves control by the infant of its environment: The infant makes an active response either in terms of vocalization or following behavior to achieve contact and proximity. In the primate social group, vocalization usually results in the mother's rapidly achieving proximity with her infant and frequently retrieving it. Therefore, the infant uses vocalization as an active signal to achieve contact and also to control certain aspects of the mother's behavior. Rosenblum and Plimpton (1981) argued that one possible function of the high rates of vocalization observed during the first day of separation may be an attempt by the infant to reestablish contact with its mother. However, if vocalization fails to achieve proximity and contact, or the stimulus that elicits vocalization (namely the mother) is not present, one would predict reduced vocalization rates. Total separation from the mother represents a situation in which there is loss of ability by the infant to achieve proximity and contact.

Social Buffering

The data that have emerged from our research indicate another important qualification, namely that the behavioral and endocrine responses following separation are dependent upon the condition of separation. In particular, social variables appear to have a potent effect on the developing infant's biobehavioral responses to separation. In another series of studies, we demonstrated that social factors can modulate both the behavior and the level of pituitary-adrenal axis activity in the infant primate faced with maternal loss. The initial study described above on the pituitary-adrenal response of infant squirrel monkeys to separation (Mendoza *et al.*, 1978) was conducted under conditions in which the mother and infant were housed as a dyad in individual cages. However, since squirrel monkeys, as well as other primates, spend most of their time in social groups, we felt impelled to examine the effects of separation in a group living situation (Coe *et al.*, 1978a). A group of squirrel monkey mothers and infants were tested as follows: The first condition involved taking a blood sample from the mother and infant immediately following removal from the group in order to determine basal cortisol concentrations. We have demonstrated that the handling involved with capture of the subjects does not result in a significant elevation in cortisol within this time frame (Coe *et al.*, 1978b). A second condition, separation-reunion, involved a momentary separation and immediate reunion of the dyad, with a blood sample taken 30 min later, in order to determine the effects of the capture procedure. The third condition involved the removal of the mother from the social group, allowing the infant to remain alone in the social group for 30 min before blood sampling.

One aspect of the social group used in this study was that it contained a late-pregnant female in addition to the mother-infant dyads. Under some circumstances female squirrel monkeys which do not have infants will show maternal behavior in that they permit the infant to ride dorsally and also attempt to retrieve the infant. This alloparenting behavior by nonlactating females is known as "aunting". The late-pregnant female in this experiment did aunt each of the infants whose mother was removed. The infants which were aunted by the pregnant female showed few signs of the usual protest behavior observed immediately following separation.

The phenomenon of reduced behavioral agitation (both activity and vocalization levels) when an infant is with an aunt has been interpreted by Rosenblum as indicating that aunting

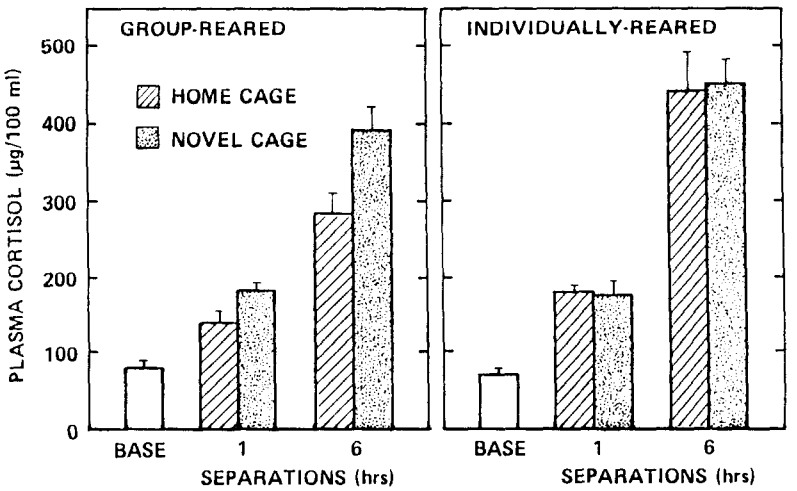


FIG. 3: Infant squirrel monkey plasma cortisol concentrations (µg/100 ml) of group-reared or individually reared dyads during nondisturbed conditions (BASE) and following separations in the home or novel cage of 1 or 6 hr. Shown are means ± SEM.

reduces the stressful effects of separation. The data obtained from this experiment indicated that, although aunting does reduce the behavioral manifestations which are observed in separated infants, these infants showed a pituitary-adrenal response that was significantly elevated over basal and separation-reunion conditions. In fact, the cortisol elevation was as high as when the infants were removed from the mother and placed in a novel cage. These isolated infants displayed high levels of vocalizations, in contrast to the relatively quiescent state of the infants which were aunted.

We interpreted these data as indicating that the infant shows specific recognition of the mother and that only the mother has the capacity to reduce arousal in the infant. However, research over the ensuing years has more clearly indicated that the presence of social partners does affect the response to separation in the infant squirrel monkey. These later studies indicate that contact with other members of the social group, and not aunting alone, can modulate the infant's response to separation from its mother. It is important to note that our original conclusions were based on blood samples obtained 30 min following separation from the mother; the modulating effect of social partners appears with longer separations. The phenomenon of "social buffering", that is, the capacity of the social group to ameliorate the behavioral and pituitary-adrenal response to separation, is clearly demonstrable when one lengthens these studies (Coe *et al.*, 1985; Wiener *et al.*, 1987). Perhaps one lesson to be learned is the difficulty in interpreting plasma cortisol data based on only two time points (i.e., a baseline sample and a single post-separation sample), rather than the evaluation of the time course of the response following the experimental manipulation. It should be noted that when infants are separated in the presence of familiar conspecifics, there is both a reduction in circulating levels of cortisol and vocalization. This apparent contradiction between our previously stated inverse relationship between rates of vocalization and cortisol levels is resolved by hypothesizing that social interaction with familiar cage mates may represent a non-vocal coping response (e.g., proximity and contact with other monkeys). Further, the

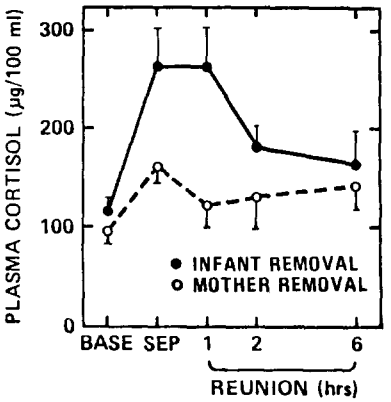


FIG. 4: Infant squirrel monkey plasma cortisol concentrations (µg/100 ml) under a nondisturbed condition (BASE), following a 30-min separation from the mother in either a novel cage (INFANT REMOVAL) or allowed to remain in its home social group after the mother was removed (MOTHER REMOVED), and 1, 2 and 6 hr following reunion with the mother. Shown are means ± SEM.

predictability of the social interaction would also act to modulate the endocrine response to separation.

In another experiment (Wiener *et al.*, 1987), infants were reared either in small maternal groups consisting of three or four mother-infant (group-reared) dyads, or only with their mothers in individual cages (individually-reared). When the infants that were reared in social groups were removed from their mothers and placed in a novel environment for one or six hours, there was a reliable elevation of plasma cortisol (Fig. 3). However, if the same infants were allowed to remain in the social home environment following removal of the mother, there was a significant diminution of the plasma cortisol response following separation. There were two important differences in this experiment compared to the earlier experiments in which infants were permitted to remain in their social groups. First is the longer lengths of separations experienced by the infant (1 or 6 hr) before blood sampling and second is that in this experiment all females in the social group had their own infants. Thus, when the mother was removed from the social group, there were no potential aunts, and aunting did not occur. However, as noted above, even in the absence of aunting, the ability of the familiar social group to ameliorate the plasma cortisol response following separation is apparent. In contrast, when the same experimental conditions were imposed upon the infants reared alone with their mothers, the resulting pattern was considerably different. The adrenal response to separation observed in the individually-reared infant allowed to remain in its familiar home cage was as great as that observed in the same infant when it was separated from its mother and placed in a novel environment. Therefore, in order to ameliorate the plasma cortisol response to separation, the critical aspect of the home environment appears to be the presence of familiar conspecifics and not familiarity with the cage itself.

Yet another example of the influence of social variables on the response to separation was demonstrated by the following experiment (Coe *et al.*, 1985). This study utilized two paradigms that we have previously employed. First was the paradigm where either the infant is removed from the social group and placed in a novel environment or the mother is removed and the infant permitted to remain with its social partners. The second was to utilize the response following reunion with the mother as a means of evaluating the impact of separation. Typically, when the infant is not reunited with the mother, there is a continuing elevation of plasma cortisol over the first day. If reunion is permitted, the infant begins to show a rapid return to normal adrenal hormone output. In this study, we examined the rate of recovery over

a 6-hr period in infants that were either separated from the mother for 30 min in a novel cage alone (infant removal) or separated for 30 min in the home social group following mother removal (mother removed) (Fig. 4). The adrenal response to separation was significantly lower when the infants were allowed to remain in the social group, and at 1 to 2 hr following reunion with the mother. In this experiment we used a 30-min separation; in contrast to our earlier report (Coe *et al.*, 1978a), the animals in this study displayed a reduction in the plasma cortisol response to separation if permitted to remain in the social group even at 30 min. In addition, behavioral observations during the 6 hr following reunion indicated that, if the infants were separated in the home cage, they showed a more rapid behavioral adjustment during the reunion period as indicated by their willingness to depart from the mother and resume play behavior. Animals who were reunited with their mothers after being placed in a novel environment tended to show a longer period of maternal contact and clinging. Following reunion, there was no further increase in infant plasma cortisol levels and a slow return to baseline. Thus, contact with the mother and the social environment terminates the pituitary-adrenal response to separation. The data once again indicate that familiarity with the physical and social environment facilitates the infant's ability to deal with separation, and that reunion with the mother initiates a process that ameliorates pituitary-adrenal endocrine activation.

Presence of Mother During Separation

The degree of access to the mother during separation is another important aspect of the separation environment that affects the infant's biobehavioral response. Suomi *et al.* (1976) reported that rhesus infants separated from their mothers by a wire mesh partition which permitted physical contact showed only minimal "distress behavior" (i.e., vocalization). However, when dyads were separated by a Plexiglas panel which prevented tactile contact, the infants vocalized a lot. Seay and Harlow (1965) reported that infants with no access to the mother, because of placement in another room, exhibited fewer vocalizations than infants separated from the mother yet in view of her through a Plexiglas window.

Early research relied heavily on vocalization as a measure of distress, and thus, it was assumed that exposure to the mother produced greater distress. Reduced vocalization rates of infants totally isolated from the mother supported the out-of-sight, out-of-mind idea. However, in view of the inverse relationship between cortisol and vocalization observed in squirrel monkeys, we prefer the hypothesis of Rosenblum and Plimpton (1981), that the initial period of vocalization represents the infant's attempt to reestablish contact. We have argued that these initial vocalizations, particularly when visual access to the mother is available, represent active coping attempts on the part of the infant. If this hypothesis has validity, then we would predict that under conditions where the infant had visual access to the mother, it would display increased vocalization and a concomitant reduction in plasma cortisol.

A series of studies conducted in our laboratory on rhesus macaque infants supports these hypotheses (Levine *et al.*, 1984; 1985). Infants separated and totally isolated from their mother showed fewer vocalizations and significant increases in plasma cortisol above basal concentrations. In contrast, infants separated but with visual access to their mothers, showed a significantly higher number of vocalizations and either a reduced cortisol response or, under some circumstances, no cortisol response at all. The vocalizations differed not only in the number emitted under these two separation conditions, but also in many of their sonic characteristics (Levine, 1987).

The pituitary-adrenal response following separation in infant squirrel monkeys and infant rhesus macaques is different. The infant squirrel monkey shows a more robust response, and of longer duration. However, similar environmental variables influence the behavioral and endocrine responses in the two primate species. It is not surprising, therefore, that separated infant squirrel monkeys placed in a cage adjacent to their mothers display similar biobehavioral

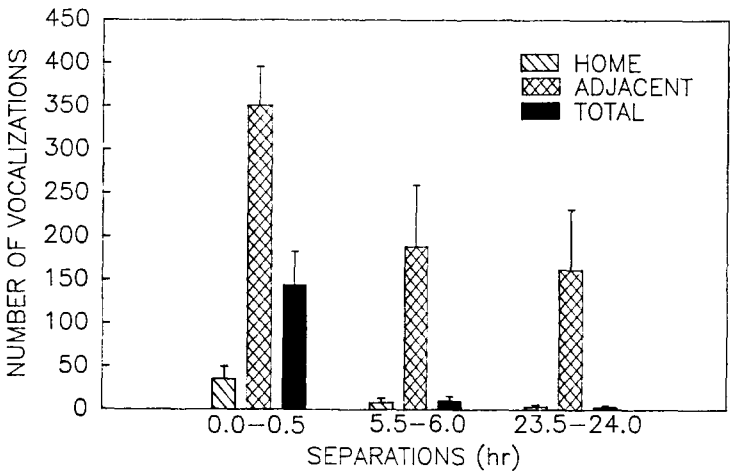


FIG. 5: Infant squirrel monkey vocalizations during the first 30 min of separation, 5.5-6.0 hr following separation, and 23.5-24.0 hr in the home social group following mother removed (HOME), isolated in a novel cage (TOTAL), or in a cage adjacent to the mother (ADJACENT). Shown are means ± SEM.

responses: a prolonged period of vocalization and a reduced elevation of plasma cortisol compared to totally isolated infants (Wiener *et al.*, in press). At the end of 24 hr, the infants separated from but adjacent to their mothers continued to vocalize, whereas the totally isolated infants emitted few, if any, vocalizations. It should be noted that the conditions of this experiment were different from those in our rhesus studies. The squirrel monkey infant was separated from its mother by an opaque barrier, and thus only auditory and olfactory communication was possible; visual and tactile access to the mother was prevented.

In another experiment conducted on squirrel monkeys we examined the behavioral and endocrine responses following three conditions of separation: (1) Adjacent, in which the infant was placed in a cage adjacent to its mother and permitted visual, olfactory and auditory communication but not tactile contact; (2) Total isolation, in which the infant was placed in a cage located in another facility away from the mother; (3) Home, in which the infant was permitted to remain in its social group following removal of the mother. Behavioral observations were conducted during the first 30 min, 5.5 to 6 hr, and 23.5 to 24 hr following separation. Plasma cortisol concentrations were determined at the end of the 24 hr. Vocalization following the first 30 min of separation showed a curvilinear relationship across the three conditions of separations (Fig. 5). The fewest vocalizations were displayed by the infants in the home condition, and the most vocalizations were shown by those infants in a cage adjacent to their mother. The totally isolated infants displayed fewer vocalizations than the infants in the adjacent condition, but more calls than those in the home condition. In contrast, the plasma cortisol response to separation showed a linear increase: Each of the three conditions was significantly greater than its less-isolated comparison (Fig. 6). Separation in the home environment, therefore, results in a reduced plasma cortisol response compared to separation in the adjacent environment, and separation in the adjacent environment results in a reduced cortisol response compared to separation under total isolation. These data indicate a differential sensitivity of the pituitary-adrenal system to different environmental events.

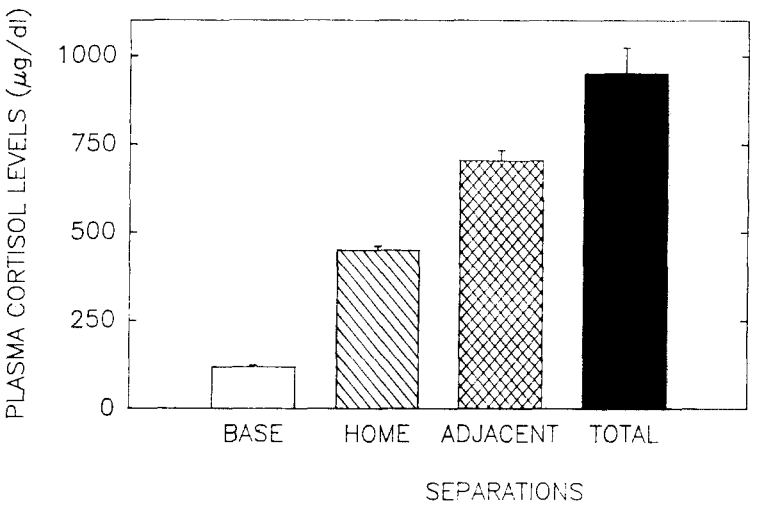


FIG. 6: Infant squirrel monkey plasma cortisol concentrations (μg/dl) during a nondisturbed home condition (BASE), 24 hr after the removal of the mother from the social group (HOME), placement in a cage adjacent to its mother (ADJACENT), or separated to an isolation novel cage (TOTAL). Shown are means ± SEM.

CONCLUSION

In conclusion, we have described several social factors which affect the plasma cortisol response during disruptions of mother-infant relationships. We believe that the effects of these social factors can be viewed within the general theory of stress and coping. One of the most potent psychological variables which activates the pituitary-adrenal system can be described as uncertainty. All situations vary on a continuum from highly predictable (certain) events to highly unpredictable (uncertain) events. Uncertainty can be evoked by novelty, by insufficient information concerning the nature of forthcoming events, or by loss of control over events which affect the organism. The pituitary-adrenal response to a novel stimulus reflects the organism's response to uncertainty, because, by definition, there is little information possessed by the organism based upon its previous experience which can be used to assess the new environment. Any information about the stimulus or cognitive operation that serves to make a situation less uncertain, whether it be passive (i.e., habituation) or active (i.e., by exerting control), should lead to a reduction or amelioration of pituitary-adrenal activity.

The availability of stable and familiar social relationships provides a set of predictable outcomes of interactions, due in part to a long history of past interactions and experiences. Thus, the predictability of the social interactions in a group of familiar conspecifics would, by definition, reduce uncertainty. However, it appears that it is not only actual social interaction involving physical contact which can reduce uncertainty, but also exposure to highly salient social stimuli (e.g., mother adjacent to infant), even in the absence of physical contact, which can provide the organism with information that results in reduced arousal and cortisol output. The availability of stable social partners provides a mechanism whereby uncertainty is reduced. It follows that, if an infant is separated from the mother and placed in an unfamiliar social group, we would predict that it would show a maximum level of uncertainty and a high level of pituitary-adrenal activation. We have tested this hypothesis by separating infant squirrel

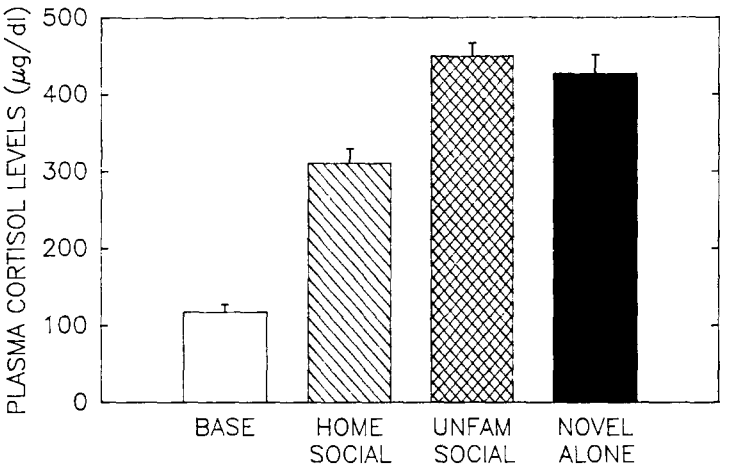


FIG. 7: Infant squirrel monkey plasma cortisol concentrations (µg/dl) during nondisturbed conditions (BASE) and following 4 hr of separation from the mother by her removal from the home social group (HOME SOCIAL), placement in an unfamiliar social group comprised of mother-infant dyads (UNFAM SOCIAL), or placed in an isolated novel cage (NOVEL ALONE). Shown are means ± SEM.

monkeys from their mothers and placing them in unfamiliar social groups. Their response was compared with their response when totally isolated (Fig. 7). The data only partially supported the hypothesis, in that the magnitude of the cortisol response of separated infants placed in unfamiliar social groups was not significantly higher than that of the totally isolated infants. It is possible, however, that we have reached the ceiling of the infant's capacity to display an adrenal cortisol response within the 4-hr separation used in this study.

The data we have presented document the importance of social variables in modulating behavioral and biological responses of the infant nonhuman primate following disturbances in the mother-infant relationship. This has lead us to concur with the view held by Kummer (1971) that one of the most important adaptations developed by higher primates is their ability to engage in social relationships involved in group living.

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